WHITE PAPER · SAMSUNG SL-45A/2200 × MILITARY & DEFENSE

Samsung SL-45A for Military & Defense

Heavy CNC lathe purpose-built for forged blanks, rough shafts, drive mandrels, and structural stock. This paper covers what the machine does, the military & defense parts we produce with it, the alloys we run, the tolerances and finishes military & defense buyers expect, and the documentation package.

Hammer of Forge · 14" OD × 96" L max turning · heavy-duty CNC lathe for rough stock

Executive summary

Machine: Samsung SL-45A/2200 (shop nickname: *Hammer of Forge*). heavy-duty CNC lathe for rough stock at B&R Productions, running from our New Waverly, Texas shop under an ISO 9001:2015 quality system.

Application: Heavy CNC lathe purpose-built for forged blanks, rough shafts, drive mandrels, and structural stock for Military & Defense customers — Defense primes, subassembly manufacturers, tactical vehicle component OEMs, munitions component suppliers, weapon-system integrators, sustainment (out-of-production replacement) contractors.

Envelope: 14" OD × 96" L max turning. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

Hammer of Forge is the machine that takes ugly starting material and makes it round, straight, and to size. Forged blanks with scale and skin, rough saw-cut billets, oversized bar stock — this machine has the rigidity and horsepower to get through the roughing pass without chatter and without walking the part. Most of what our other lathes finish, this one starts.

Full spec table

Model	Samsung SL-45A/2200 ("Hammer of Forge")
Type	Heavy-duty CNC turning center — Samsung Machine Tools SL series
Max turning capacity	14" outer diameter × 96" length
Chuck	Large 3-jaw chuck sized for forged stock
Rigidity	Samsung SL-A series designed for rough-turning stability
Tailstock	Full-length tailstock support for slender long parts
Tolerance capability	±0.0005" routine after roughing; excellent for finish passes
Materials	Forgings, castings, bar stock — 4140/4340, stainless, Inconel, Duplex, 17-4 PH

Who this serves in Military & Defense

Typical buyer: Defense primes, subassembly manufacturers, tactical vehicle component OEMs, munitions component suppliers, weapon-system integrators, sustainment (out-of-production replacement) contractors.

What's hard about military & defense work: ITAR compliance, security-sensitive prints, program-specific quality plans, tight-tolerance structural components, exotic-alloy selection driven by service environment, and legacy-part sustainment where the OEM no longer supports the design.

Typical Military & Defense parts we produce on Hammer of Forge

- Rough-turning of defense forgings
- Structural round parts starting from forgings
- Weapon-component blanks and roughing pass
- Vehicle-component large-diameter roughing
- Roughing to prep parts for finish on precision machines

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Military & Defense

Standard range: 17-4 PH · 15-5 PH · 13-8 Mo · 4140 pre-hard · 4340 · Titanium Grade 5 · Inconel 718 · Aluminum 7075 · High-strength steels · Customer-specific alloys on request.

Defense parts routinely spec 4140 pre-hard, 4340, and precipitation-hardening stainless — condition-specific machining knowledge matters, same as aerospace. Titanium is common on weight-critical structural components. Legacy sustainment often specifies materials nobody sees anywhere else; we work with the spec sheet you provide and validate machining strategy before we cut.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

Tolerance and finish expectations in Military & Defense

±0.0005" routine on critical features. Tighter with proper setup and fixture. GD&T-driven tolerancing supported. CMM verification on first articles; customer-plan-driven measurement on production runs. Documentation the level a program office expects.

Documentation and quality workflow

ITAR-controlled work under NDA with customer-specific document control — prints stay in-shop. Customer-specific quality plans, PPAP, and first-article documentation standard. Material traceability by heat and lot on every job. Certificates of conformance with delivery.

Response time and emergency work

Program-driven schedules honored. Emergency component replacement supported for sustainment programs with established relationships. Legacy-part reverse-engineering supported for out-of-production sustainment.

How we compare

Compared to a large defense-prime captive shop: we're faster for one-offs, small runs, and legacy sustainment where the prime's supply chain has moved on. Compared to a lowest-bidder commodity shop: we handle ITAR, we document properly, and we understand that a program-office audit is a real thing.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request. ITAR-controlled prints handled under document control with prints staying in-shop.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Ready to talk about your Military & Defense work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

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